

AIR BALANCE SCHEDULE		
EQUIPMENT	EXHAUST CFM	OUTDOOR AIR CFM
EF-1 (4x150 CFM EA.)	600	
EF-2 (2x150 CFM EA.)	300	
KH-1	600	
RTU-1		1500
TOTAL	1500	1500

FAN SCHEDULE										
TAG	MANUFACTURER MODEL	FAN TYPE	CONT. CFM	INT. CFM	S.P.	FAN RPM	SONES AT OPER. PT.	ELECTRICAL		ADDITIONAL FEATURES REQ.
								HP	VOLTS/PHASE	
EF-1	PANASONIC FV-0511VKS2	CEILING DIRECT - TWO SPEED	50	150	0.125	935	0.4	9 WATTS	115/1	1,3,5,9
EF-2	GREENHECK SP-B150	CEILING DIRECT	-	150	0.25	1050	3	153 WATTS	115 / 1	1,3,5,8
EF-3	GREENHECK SP-A410	CEILING DIRECT	-	300	0.25	1000	3	121 WATTS	115 / 1	1,3,5,15
EF-4	BY OWNER	PLYMOVENT UTILITY	-	1000	6	3495	-	3	208 / 3	BY OWNER
GEF-1	GREENHECK SE1-12432-A4	SIDEWALL PROPELLER	-	500	0.25	1750	28	1/4	115 / 1	1,6,7,10
GEF-2	GREENHECK SBE-1H24-4	SIDEWALL PROPELLER	-	3000	0.25	923	16.7	1/4	115 / 1	1,6,7,11,12,13,17
TF-1	GREENHECK CSP-A125	TRANSFER FAN	-	100	-	980	0.3	44 WATTS	115 / 1	1,10
CF-1	AIRIUS AIR PEAR A-25-EC	DESTRATIFICATION CIRC FAN	-	620	-	1640	69.1 Dba	31.3 WATTS	115 / 1	14
FEATURES: 1. CORRECT ALL FAN RPM FOR ALTITUDE. 2. SWITCH ON WALL. 3. FLAT ROOF CAP OR BLACK WALL CAP. 4. PROVIDE WITH 14" ROOF CURB. 5. GRAVITY BACKDRAFT DAMPER. 6. SHORT WALL HOUSING W/ GUARD AND MOTORIZED DAMPER. 7. GALVANIZED DAMPER GUARD. 8. CONTROL WITH WALL MOUNTED SPEED CONTROLLER. 9. TWO SPEED OPERATION. CONTINUOUS OPERATION AT LOW SPEED W/ WALL SWITCH FOR HIGHER AIRFLOW. 10. FAN SHALL RUN CONTINUOUSLY. 11. CONTROL WITH CARBON MONOXIDE/NOX SYSTEM. 12. INTERLOCK WITH OA INTAKE HOOD MOTORIZED DAMPER. 13. PROVIDE MOTORIZED DAMPER. 14. VARIABLE SPEED 5A CONTROLLER. OFF WHITE COLOR. 15. CONTROL WITH WALL SWITCH VT20M. CONTINUOUS AT 100 CFM W/ 12 HR. (ADJ.) TIMER. 16. CONTROL WITH REVERSE-ACTING THERMOSTAT. SET AT 80° (ADJ). 17. PROVIDE MANUAL SWITCH TO OPERATE DURING SUMMER HRS 18. FAN IS ACTIVATED BY TIMER RELAY. EQUALS BY: SEE SPECIFICATIONS										

ELECTRIC UNIT HEATER SCHEDULE								
TAG	MANUFACTURER	MODEL	TYPE	CFM	KW	ELECTRICAL		ADDITIONAL
						HP	VOLTS/PHASE	FEATURES REQUIRED
EUH-1	MODINE	HER 30	HORIZ.	380	3	1/40	208 / 1	1,2
FEATURES: 1. UNIT MOUNTED T'STAT. 2. HANGER KIT.								

RADIANT HEATER SCHEDULE							
TAG	MFGR	MODEL	TYPE	LENGTH (FT)	S.L. FUEL INPUT MBH	VOLTS/PH.	ADDITIONAL FEATURES REQUIRED
IRH-1	ROBERTS GORDON	CTH2V-125	NG	35	125	120/1	1, 2, 3
FEATURES: 1. PROVIDE WITH ROOF COMB. AIR & VENT. 2. PROVIDE WITH MFG. HANGER KIT. 3. PROVIDE WITH A REDLINK ENABLED HONEYWELL H8321R001 THERMOSTAT. NO EQUALS.							

G.R.D. AND LOUVER SCHEDULE						
TAG	MFG / MODEL		FIRE DMPR	OBD	MAX N.C.	ADDITIONAL FEATURES REQUIREMENTS
A	PRICE SMDA	LOUVERED ADJUSTABLE SUPPLY DIFFUSER	--	YES (G)	30	STEEL VERTICAL DEFLECTION
B	PRICE B2	EGGCRATE RETURN GRILLE	--	YES (F)	30	ALL ALUMINUM 1" X 1" X 1" CORE
D	PRICE 520S	SIDEWALL SUPPLY	--	YES	30	ALL STEEL DOUBLE DEFLECTION 3/4" SPACING
F	PRICE SDGE	SPIRAL SIDEWALL SUPPLY	--	YES	30	ALL ALUMINUM DOUBLE DEFLECTION 3/4" SPACING. RADIUS END CAP TO MATCH DUCT RADIUS WITH DAMPER EXTRACTOR.
S	PRICE SDS OR SDB	SLOTS SEE PLANS FOR # SLOTS/WIDTH/LENGTH@ DUCT DIA	--	NO	30	PROVIDE 1/2" INSULATED PLENUM W/ OUTLET SIZE SHOWN ON PLANS. PROVIDE TYPE 2 CONCEALED FRAME IN HARD CEILINGS. PROVIDE TYPE 15 LAYON IN T-BAR CLGS. PROVIDE TYPE 16 CURVED TO MATCH SPIRAL DUCT RADIUS W/EQUALIZING CORE IN SPIRAL DUCT.
T	PRICE ATG1C	DOOR GRILLE. TRANSFER	NO	NO	30	ALL ALUMINUM
L	GREENHECK ES4401	LOUVER - EXHAUST OR COMBUSTION AIR	NO	NO	--	ALL ALUMINUM WITH EXTRUDED FRAME. BLACK FINISH, AMCA 0.01 OZ. AT 750 FT.MIN.
M	GREENHECK ESD 435	LOUVER - INTAKE AIR DRAINABLE	NO	NO	--	ALL ALUMINUM WITH EXTRUDED FRAME. BLACK FINISH, AMCA 0.01 OZ. AT 1000 FT.MIN.
FEATURES: A. USE LAY-IN FRAME STYLE 3P ON ALL T-BAR CEILINGS. B. USE FRAME STYLE 1 ON ALL HARD SURFACE CEILINGS. C. USE MAX 6FT. OF CODE APPROVED INSULATED FLEX DUCT. D. MAXIMUM S.P. DROP = 0.15" W.C. UNLESS OTHERWISE NOTED. E. ALL CEILINGS DIFFUSERS TO HAVE 4-WAY DEFLECTION UNLESS SHOWN WITH THROW BLOCKING. F. OBD MAY BE OMITTED IF ONLY ONE RETURN INLET PER SYSTEM IS USED, OR RETURN SYSTEM IS NON-DUCTED. G. USE SPIN-IN FITTINGS WITH LOCKING BUTTERFLY DAMPER IN ACCESSIBLE LAY-IN CEILINGS. H. IN LIEU OF OBD. USE OBD IN ALL NON ACCESSIBLE CEILING AREAS ONLY WHERE SPIN/DAMPER CAN NOT BE SERVICED. EQUALS BY: GRD - KRUEGER, NAILOR, ANEMOSTAT; LOUVERS - RUSKIN, ARROW, UNITED ENERTECH NOTE: PROVIDE A ROOM-BY-ROOM AIR DISTRIBUTION SCHEDULE WITH THE DIFFUSER AND GRILLE SUBMITTAL. INCLUDE TAG #, ROOM #, MANUFACTURER, MODEL #, NECK SIZE, BORDER SIZE, COLOR, OBD, QUANTITY, CFM AND N.C. THROW @ 150 FPM. MAX AIR FLOW @ DUCT & FLEX 0.15" PER 100' SIZE CFM 6₆ 130 7₆ 200 8₆ 300 10₆ 500 SIZE CFM 12₆ 800 14₆ 1200 16₆ 1800 18₆ 2400						

ROOFTOP HEATING & COOLING UNIT SCHEDULE (HIGH EFFICIENCY W/410A REFRIGERANT)																									
TAG	CARRIER MODEL	COOLING			HEATING MBH			SUPPLY FAN @ ALTITUDE				Power Exhaust					Height	DIMENSIONS		VOLTS / PHASE	UNIT MCA	UNIT MOC	UNIT WT (LB)	ADDITIONAL FEATURES REQUIRED	
		ARI MBH	5300 FT ALT MBH	EER / IEER	SEA LEVEL INPUT	GAS CFH	5300 FT ALT. OUTPUT	TOTAL CFM	MIN OA CFM	ESP	RPM	FAN BHP	CFM	ESP	HP	MCA	MOC	Unit with 24" Curb	LENGTH						WIDTH
RTU-1	48LCRB14F3M5	146.0	121.9	12.2	192/240	192/240	118.0/148.2	4800	1500	1.50	1080	3.7	4700.0	0.75	2.0	8.1	14.6	6'-10.5"	11'-9.5"	7'-2.4"	460/3	36.9	50	2200	1,2,4,5,6,7,8,10,12, 14,15
FEATURES:																									
1. STANDARD BELT DRIVE FAN WITH VARIABLE PITCH MOTOR SHEAVE RATED AT 1.15 X MOTOR HP.													12. CONDENSER COIL HAIL GUARDS.										EQUALS BY: SEE SPECIFICATIONS		
2. FACTORY ROOF CURB (SHIM AS REQUIRED TO LEVEL).													13. CO2 REMOTE WALL MOUNTED CONTROL OF MINIMUM OUTSIDE AIR.												
3. (MANUAL/2-POS MOTORIZED) FRESH AIR HOOD 0-100%.													14. RETURN SMOKE DETECTOR WITH SAMPLING TUBE AND REMOTE TEST STATION WITH (AUDIBLE/VISUAL) ALARM												
4. NON-FUSED DISCONNECT BY E.C. COORDINATE W/ E.C.													SMOKE DETECTOR SHALL BE CONNECTED TO THE FIRE ALARM SYSTEM AND REPORT AS SUPERVISORY SIGNAL AT FIRE ALARM CONTROL PANEL.												
5. POWERED 120V/160 CONVENIENCE OUTLET.													15. MICROMETL MODULATING POWER EXHAUST TO MATCH RTU. PROVIDE FIELD SUPPLIED DISCONNECT. (POWER EXHAUST MCA, MOC & WT. INCLUDED IN UNIT TOTALS).												
6. ULTRA LOW LEAK ECONOMIZER W/ BAROMETRIC RELIEF AND FAULT DETECTION.													GENERAL NOTES (REQUIRED ALL UNITS):												
7. FILTER SECTION W/ 2 SETS OF 2" MERV 8 FILTERS.													1. ALL EQUIPMENT SHALL COMPLY WITH LOCAL ENERGY CODE REQUIREMENTS AND ASHRAE 90.1 - 2016.												
8. STANDARD INDUCED DRAFT, TUBULAR, ALUMINIZED STEEL HEAT EXCHANGER.													2. HEATING OUTPUT CAPACITY BASED ON 5300 FT ALTITUDE.												
9. 409 STAINLESS STEEL HEAT EXCHANGER.													3. CAPACITY IS BASED ON CFM @ ESP AND ELEVATION SHOWN AND 78 DB / 53 WB EAT AND 90 DEG F AMBIENT CONDENSER AIR												
10. RTU TO COME WITH BASIC ELECTROMECHANICAL CONTROLS ONLY, FIELD PROVIDED DD CONTROLLER BY T.C.C.													4. OPERATING WEIGHT INCLUDES 24" ROOF CURB, ECONOMIZER, AND HAIL GUARD.												
11. THERMOSTAT 7 DAY PROGRAMMABLE W/ LOCKING COVER.																									

PARALLEL FAN POWERED TERMINAL UNIT SCHEDULE													
TAG	MANUFACTURER MODEL	INLET SIZE	CFM			**MAX P.D.	MAX N.C.	ELEC. MOTOR VOLTS/PHASE	MOTOR HP	MOTOR AMP	ELEC. HEAT VOLTS/PHASE	ELECTRIC COIL (KW)	ADDITIONAL FEATURES REQ'D.
			MAX.	MIN.	ACTUAL								
FPB-1	CARRIER 45MD3014102	10	1440	550	920	0.75	30	240 / 1	1/4	2.8	240 / 1	5	1.2
FPB-2	CARRIER 45MD3014102	10	1440	550	1200	0.75	30	240 / 1	1/4	2.8	240 / 1	5	1.2
FPB-3	CARRIER 45MD3014102	10	1440	550	1100	0.75	30	240 / 1	1/4	2.8	240 / 1	5	1.2
FPB-4	CARRIER 45MD3015122	12	2140	1100	1500	0.75	30	240 / 1	1/2	4.6	240 / 1	6	1.2
FEATURES: 1. INTEGRATE SENSORS WITH RTU. 2. PROVIDE OPTIONAL POWER FUSE TO THE ELECTRIC HEATER.													

DUCTLESS SPLIT SYSTEM SCHEDULE (R-410A)																		
INDOOR UNIT							OUTDOOR UNIT											
TAG	CARRIER MODEL	COOLING		S.P.	CFM	UNIT WEIGHT	TAG	CARRIER MODEL	COOLING		HEATING		# OF REFRIG CIR.	ELECTRICAL (SINGLE POINT)			WEIGHT (LB)	ADDITIONAL FEATURES REQUIRED
		TOTAL MBH	SENSIBLE MBH						ARI MBH	SEER	ARI MBH	COP		MCA	MOC	VOLT/PH		
AC-1	40MAQB1B8—3	18	18.0	0.15	450	30	ACCU-1	38MAQ018	18	16	N/A	N/A	1	15	25	208-230 / 1	115	1,2,3,4,5,6,7
FEATURES:																		
1. WALL MOUNTING BRACKETS.																		
2. WIRELESS T-STAT CONTROL.																		
3. INDOOR FAN SPEEDS.																		
4. ANTICYCLE RELAY.																		
5. OUTDOOR UNIT HAIL GUARD.																		
6. REFRIGERANT LINE SETS.																		
7. MOUNT CONDENSING UNIT ON 4" CONCRETE PAD.																		
GENERAL NOTES:																		
1. INDOOR COOLING CAPACITIES BASED ON 80DB/67WB AND 95F AMB @ S.L.																		
EQUALS: SEE SPECIFICATIONS																		

KITCHEN EXHAUST HOOD W/ FAN										
TAG	MANUFACTURER	MODEL	SIZE	MAX. TEMP	CFM	DUCT OUTLET	LIGHTS	AMPS	VOLTS/ PHASE	REQUIRED FEATURES
KH-1	BLUESTAR	BS-MANH-I-54-SS	54"x34"	600°F	600	10" ROUND	HALOGEN	10	120/1	1,2,3,4,5,6
FEATURES: 1. SEE ARCHITECTURAL DRAWINGS FOR SPECIFICATIONS. 2. TYPE II STAINLESS STEEL CONSTRUCTION. CONTRACTOR TO COORDINATE AND PROVIDE ALL REQUIRED STAINLESS STEEL HOOD EXTENSIONS, BLOCKING AND SUPPORT FOR HOOD INSTALLATION AND INDICATED CEILING HEIGHT AND TYPE. 3. PROVIDE INTERLOCK WITH EO SWITCH. 4. INTERLOCK EXHAUST HOOD FAN WITH RANGE OPERATION (WHEN RANGE IS ON, HOOD IS ON). 5. MULTI-SPEED VENTILATOR CONTROL. 6. INTERLOCK WITH RTU-3 & RTU-4 FOR MAKE-UP AIR WHEN HOOD IS ON.										

GRAVITY INTAKE HOOD										
TAG	MANUFACTURER	MODEL	THROAT SIZE	CFM	DIMENSIONS			MAX. S.P.	MATERIAL	REQUIRED FEATURES
					L	W	H			
GIH-1	GREENHECK	FGI	20"x20"	3000	39"	37"	29.75"	0.25	GALVANIZED	1,2,3
FEATURES: 1. 12" HIGH BASE W/12" ROOF CURB 2. MOTORIZED DAMPER & BIRDSCREEN 3. 2" ALUMINUM FILTERS										

GARAGE VENTILATION CALCULATIONS					
FLOOR	GARAGE AREA (SF)	MIN. VENTILATION PER 0.05 CFM/SF (CFM)	FAN	SYSTEM CAPABILITY PER 0.75 CFM/SF (CFM)	FAN
APPARATUS BAY, WORK RM, HOSE TOWER, MEZZ	3,920	196	EF-4	2,940	EF-5

MINIMUM OUTSIDE AIR VENTILATION RATES (PER 2018 IMC)																		
ROOM DESIGNATIONS	OCCUPANCY CLASSIFICATION	HVAC SYSTEMS	OCCUPIED SQ. FT. (A _z)	OCC DENSITY # / 1000 SF	OCCUPANT # (P _z)	OCC VENT RATE - CFMP (R _p)	OCCUPANT CFM (R _p P _z)	AREA VENT RATE - CFMSF (R _a)	AREA VENT CFM (R _a A _z)	TOT ROOM VENT - CFM (V _{tz})	ZONE DISTRIB EFFICIENCY TABLE 403.3.1.2 (E _z)	ZONE OUTDOOR AIR FLOW - CFM (V _{oz})	PRIMARY AIR FLOW - CFM (V _{pz})	PRIMARY OUTDOOR AIR FRACTION (Z _p)	SYSTEM VENTILATION EFFICIENCY (E _v)	OUTDOOR AIR INTAKE AIR FLOW RATE - CFM (V _{vt})	ACTUAL OUTDOOR AIR INTAKE AIR FLOW RATE - CFM	COMMENTS
ENTRY 100	CORRIDORS	FPB-1	122	0	0	0	0	0.06	8	8	0.8	10	180	0.06	0.9	11	180	
DUTY OFFICE 101	OFFICE	FPB-1	260	5	2	5	10	0.06	16	26	0.8	33	450	0.07	0.9	37	450	
TOILET 102	TOILETS	FPB-1	65	0	0	0	0	0	0	0	0.8	0	50	0.00	0.9	0	50	
OFFICE 103	OFFICE	FPB-1	154	5	1	5	5	0.06	10	15	0.8	19	240	0.08	0.9	21	240	
TOTAL		FPB-1	601		3		15		34	49		62	920	0.08		69	920	
DINING 105	DINING	FPB-2	210	70	15	7.5	113	0.18	38	151	0.8	189	400	0.47	0.6	315	400	
KITCHEN 106	KITCHEN	FPB-2	402	20	9	7.5	68	0.12	49	117	0.8	147	400	0.37	0.6	245	400	
DAY ROOM 107	DAY ROOM	FPB-2	548	30	17	5	85	0.06	33	118	0.8	148	400	0.37	0.6	247	400	
TOTAL		FPB-2	1160		41		266		120	386		484	1200	0.47		807	1200	
LAUNDRY 109	COIN OPER. LAUNDRY	FPB-3	94	20	2	7.5	15	0.06	6	21	0.8	27	150	0.18	0.9	30	150	
TOILET 111	TOILETS	FPB-3	65	0	0	0	0	0	0	0	0.8	0	50	0.00	0.9	0	50	
BUNK ROOM 112	DORMITORY SLEEPING	FPB-3	98	20	2	5	10	0.06	6	16	0.8	20	120	0.17	0.9	22	120	
BUNK ROOM 113	DORMITORY SLEEPING	FPB-3	98	20	2	5	10	0.06	6	16	0.8	20	110	0.18	0.9	22	110	
BUNK ROOM 114	DORMITORY SLEEPING	FPB-3	98	20	2	5	10	0.06	6	16	0.8	20	100	0.20	0.9	22	100	
BUNK ROOM 115	DORMITORY SLEEPING	FPB-3	98	20	2	5	10	0.06	6	16	0.8	20	100	0.20	0.9	22	100	
BUNK ROOM 116	DORMITORY SLEEPING	FPB-3	98	20	2	5	10	0.06	6	16	0.8	20	100	0.20	0.9	22	100	
BUNK ROOM 117	DORMITORY SLEEPING	FPB-3	98	20	2	5	10	0.06	6	16	0.8	20	100	0.20	0.9	22	100	
CORRIDOR 118	CORRIDORS	FPB-3	360	0	0	0	0	0.06	22	22	0.8	28	220	0.13	0.9	31	220	
TOILET 119	TOILETS	FPB-3	65	0	0	0	0	0	0	0	0.8	0	50	0.00	0.9	0	50	
TOTAL		FPB-3	1172		14		75		64	139		175	1100			194	1100	
FITNESS 120	HEALTH CLUB	FPB-4	905	10	10	20	200	0.06	55	255	0.8	319	1500	0.21	0.9	354	1500	
TOTAL		FPB-4	905		10		200		55	255		319	1500	0.21		354	1500	
NOTES:																		
1. ABOVE IS BASED ON 2018 IMC CHAPTER 4																		

BOILER SCHEDULE																	
TAG	MANUFACTURER MODEL	TYPE	FUEL	% EFFICIENT	CAPACITY (MBH)		MAX. WORKING PRESSURE	FLUE/INTAKE SIZE	GPM	TEMP.		ELECTRICAL (INCLUDES P-1)			OPERATING WT. LBS.	ADDITIONAL FEATURES REQ.	
					S.L. INPUT	ALT. OUTPUT				EWI (°F)	LWT (°F)	VOLT / PHASE	MCA	MOP			
B-1	LAARS NTH 850	HIGH EFFICIENCY CONDENSING	NATURAL GAS	96	850.0	691.0	160	6" / 4"	59.2	135	110	120 / 1	15	27	600	1,2,3,4,5,6,7,8,9,10,11	
FEATURES: 1. CONTROL TYPE: MODULATING ELECTRONIC OUTDOOR RESET 2. ELECTRONIC INTERMITTANT IGNITION. 3. FLOW SWITCH. 4. GLYCOL 50%. 5. COMBUSTIBLE FLOOR PAN. 6. LOW WATER CUT OFF. 7. OUTDOOR RESET CONTROL WITH ADJUSTABLE AMBIENT LOCK OUT. 8. FLUE TYPE: DIRECT VENT (CPVC) 9. CONDENSATE NEUTRALIZER. 10. CONCENTRIC VENT KIT. 11. BOILER CIRC. PUMP INCLUDED W/ BOILER.																	
																	EQUALS BY: LOCHINVAR PATTERSON-KELLY RBI



ALLRED & ASSOCIATES

ARCHITECTURE | PLANNING | INTERIOR DESIGN

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BID SET

THORNTON, COLORADO 80602

CITY OF THORNTON

FIRE STATION #8

15343 MONACO STREET

THORNTON, COLORADO 80602

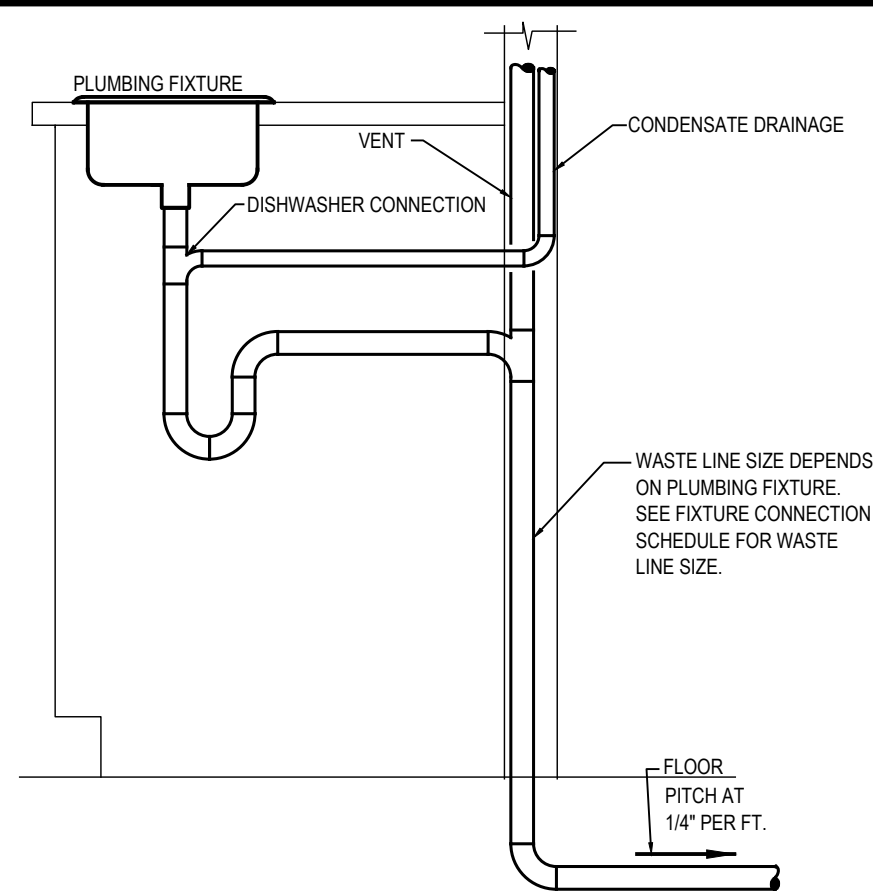


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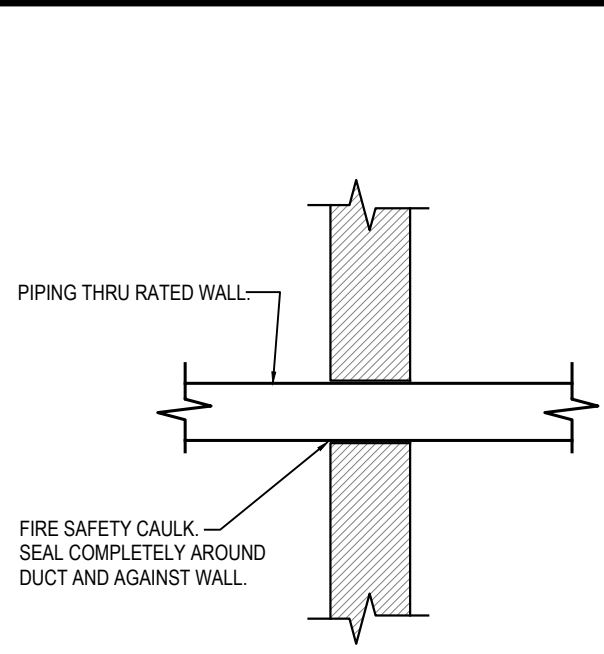
MECHANICAL
DETAIL SHEET

M1.2



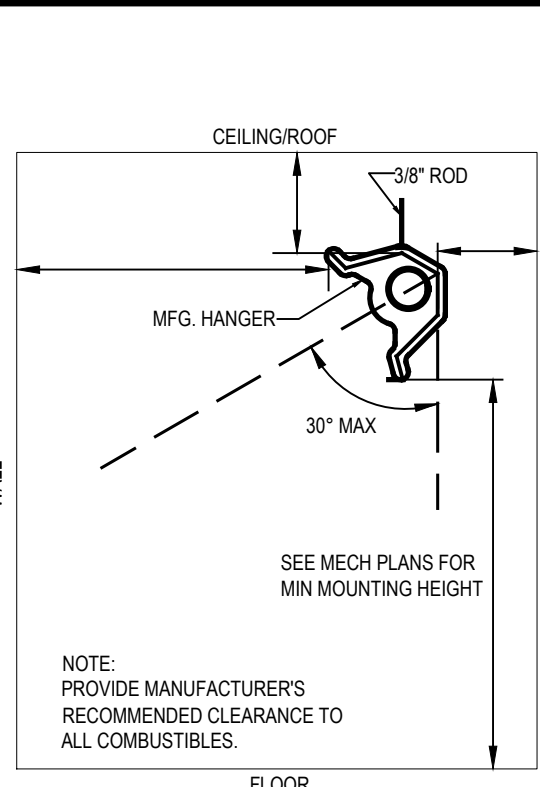
CONDENSATE DRAINAGE
CONNECTION DETAIL

NO SCALE



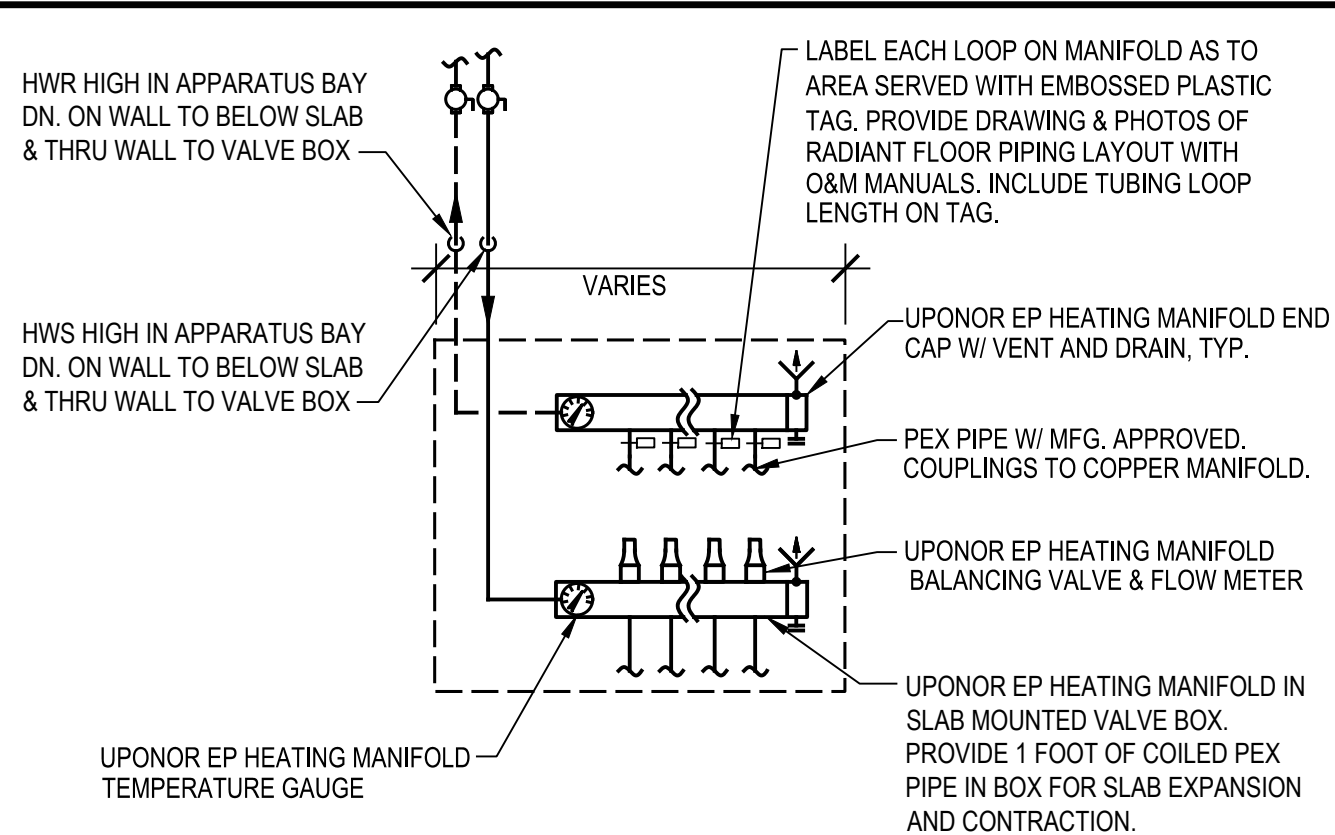
PIPE PENETRATION OF
FIRE RATED WALL

NO SCALE



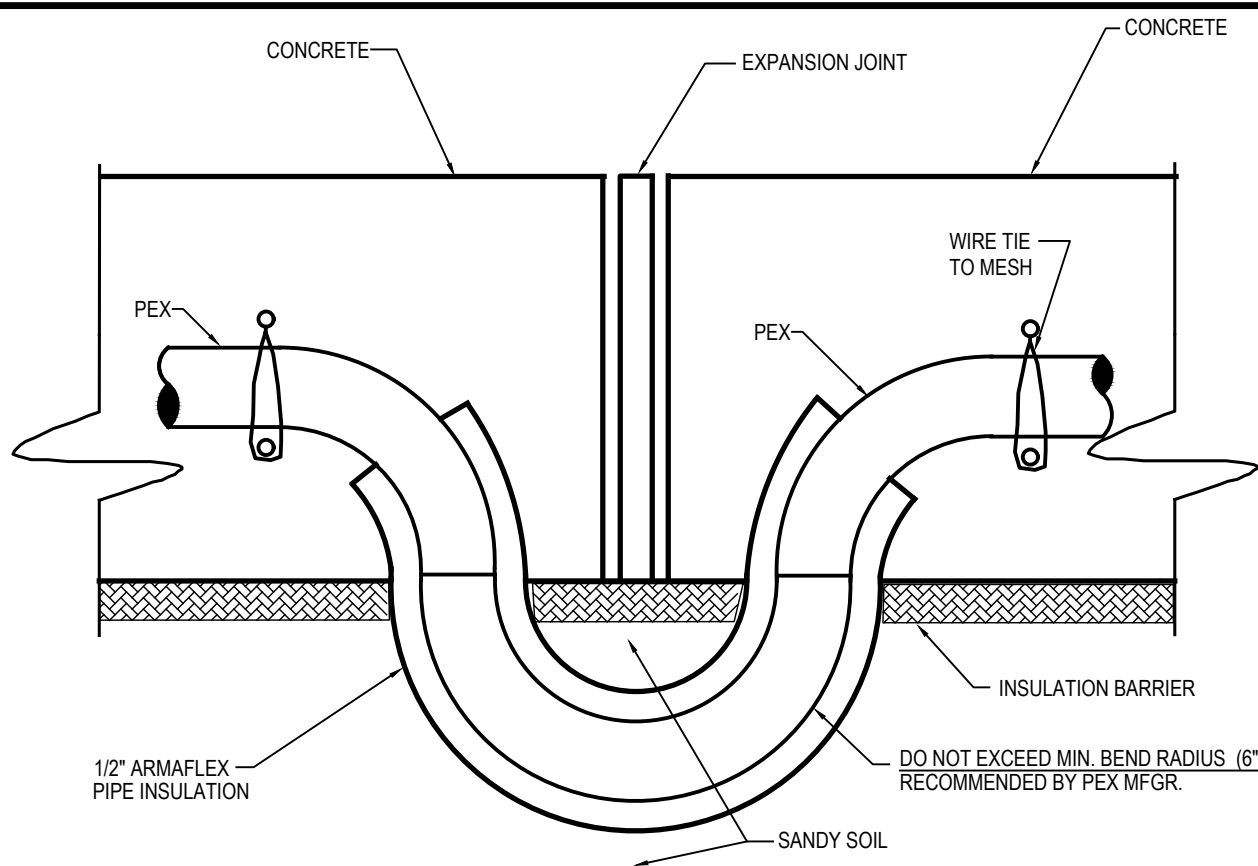
INFRA-RED HEATER

NO SCALE



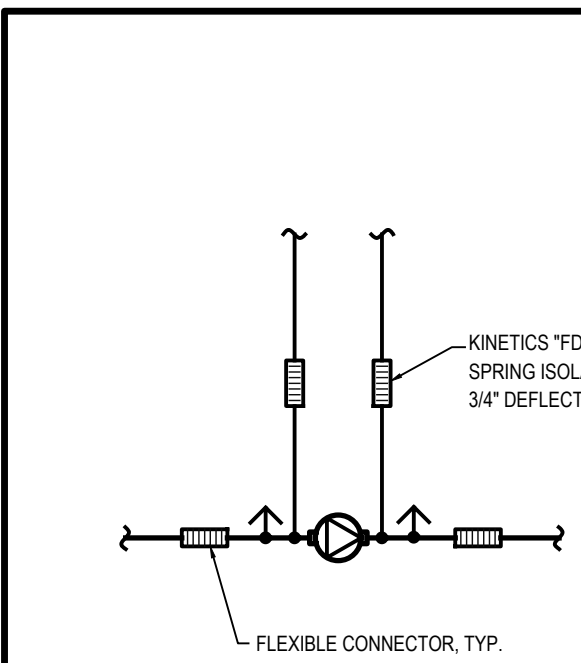
SNOWMELT MANIFOLD SCHEMATIC

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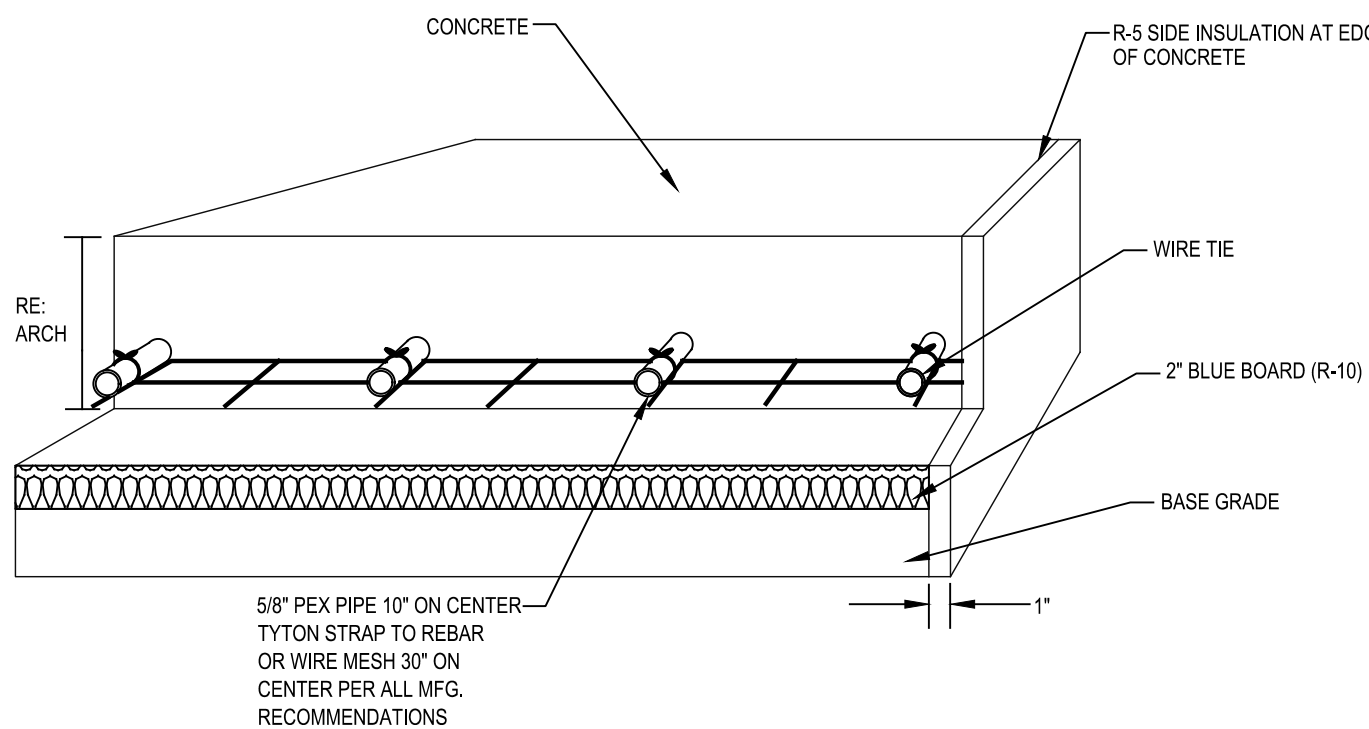
EXPANSION JOINT DETAIL

NO SCALE



PUMP SUPPORT AND
ISOLATION DETAIL

NO SCALE



SNOW MELT INSTALLATION DETAIL

NO SCALE

PUMP SCHEDULE

TAG	MANUFACTURER MODEL	LOCATION/DUTY	FLOW (GPM)	HEAD (FT)	RPM	H.P.	ELECT (VOLT/PH)	REMARKS & FEATURES REQ'D
P-2	GRUNDFOSS CR 20-1 A-GJ-A-E-HQDE	MEZZANINE / PRIMARY PUMP	60	60	1750	3.0	208 / 3	2,3,4

FEATURES REQUIRED:
1. PROVIDE LEAD-LAG SEQUENCING.
2. NON-OVERLOADING MOTOR SELECTION.
3. MULTISTAGE MOTOR.
4. SUITABLE AND SELECTED FOR 50% SOLUTION OF INHIBITED PROPYLENE GLYCOL.

EQUALS BY:
BELL & GOSSETT
TACO
ARMSTRONG

SNOWMELT HEATING ZONE/MANIFOLD LOOP SCHEDULE

MANIFOLD ZONE TAG	ZONE TAG	ZONE SNOWMELT COVER	NO LOOPS	LOOP NET SNOWMELT AREA	EWI	LWT	BTU/SF	TOTAL BTU / LOOP	TOTAL BTU	GPM	LOOP SPACING
MANIFOLD #1	ZONE 1	825	5	165	135	110	150	24750	123750	11.3	10" O.C.
MANIFOLD #2	ZONE 2	825	5	165	135	110	150	24750	123750	11.3	10" O.C.
MANIFOLD #3	ZONE 3	825	5	165	135	110	150	24750	123750	11.3	10" O.C.
MANIFOLD #4	ZONE 4	825	5	165	135	110	150	24750	123750	11.3	10" O.C.
TOTAL		3300						495000		45.1	

GENERAL
1. ALL LOOPS TO BE 5/8" PEX WITH OXYGEN DIFFUSION BARRIER.
2. ALL LOOPS TO SUPPLY ONLY GENERAL SUBLOOP AREA DESCRIBED.
3. MAXIMUM LOOP LENGTH IS 300 FEET.
4. SEE SNOWMELT PIPING DETAIL FOR INSTALLATION REQUIREMENTS.
5. MARK ALL HEADERS WITH EACH LOOP AREA SERVED.
6. ALL TUBING TO BE 10 INCHES ON CENTER EXCEPT WHERE SHOWN DIFFERENT ON DRAWINGS OR SCHEDULE.
7. TEKMAR MODEL 665 SNOWMELT CONTROLLER WITH SLAB AND OUTSIDE AIR SENSOR.

EXPANSION TANK SCHEDULE

TAG	MANUFACTURER MODEL	LOCATION	SERVICE	TYPE	ACCEPTANCE VOLUME (GALLONS)	TOTAL VOLUME (GALLONS)	ADDITIONAL FEATURES
ET-1	AMTROL AX-60V	MECH ROOM 128	SNOWMELT	DIAPHRAGM	11.3	33.6	1

FEATURES:
1. 50% GLYCOL.

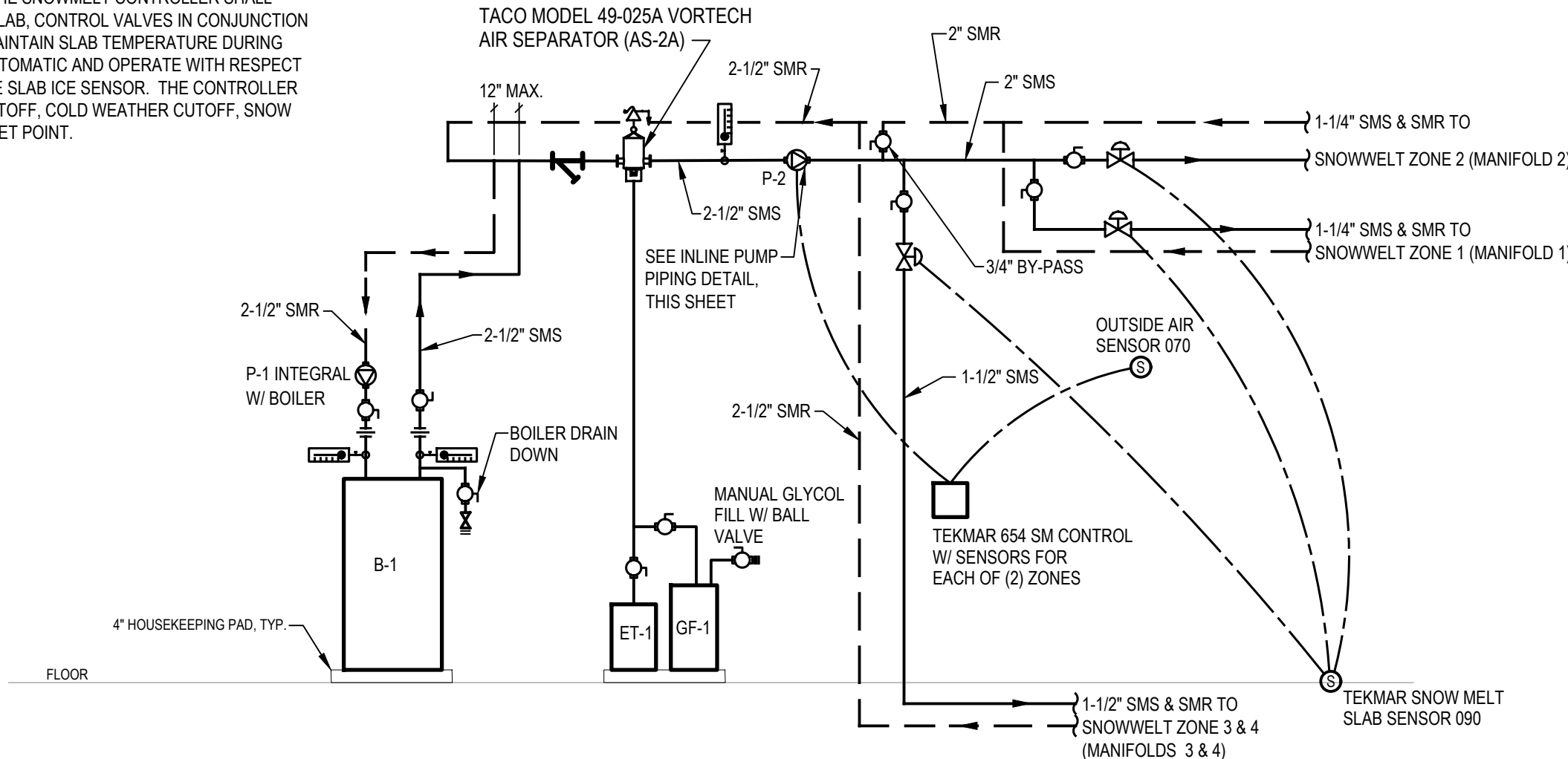
GLYCOL FEEDER SCHEDULE

TAG	MANUFACTURER	MODEL	GPM @ 50 PSIG	MAX LIQUID TEMP	ELECTRICAL		GALLONS	WEIGHT	ADDITIONAL FEATURES
GF-1	AXIOM	SF100	1	170 F	115/1	0.7	55	35 LBS	1

FEATURES:
1. PROVIDE ALARM PANEL (MODEL RA10-1). TOTAL HP WITH PUMP, FEEDER, ALARM PANEL = 1/4 HP. NEOPRENE CABEL, TYPE SJOW NO. 16/2. PANEL DIMENSIONS: 6" X 4.75" X 2.25".

SNOW MELT SYSTEM OPERATION:

THE TEKMAR 654 SNOW DETECTOR AND MELTING CONTROL FOR FOUR ZONES. P-1 (BY BOILER MANUFACTURER) SHALL ENERGIZE WHEN BOILER RECEIVES A CALL FOR HEAT FROM BOILER CONTROLLER. THE SNOWMELT CONTROLLER SHALL CONTROL P-2, WHICH CIRCULATES THE SLAB, CONTROL VALVES IN CONJUNCTION WITH THE BOILER CONTROL PANEL TO MAINTAIN SLAB TEMPERATURE DURING OPERATION. THE CONTROL SHALL BE AUTOMATIC AND OPERATE WITH RESPECT TO THE OUTSIDE TEMPERATURE AND THE SLAB ICE SENSOR. THE CONTROLLER SHALL ALSO HAVE A WARM WEATHER CUTOFF, COLD WEATHER CUTOFF, SNOW MELTING SET POINT, AND SNOW IDLING SET POINT.



SNOWMELT PIPING DIAGRAM

NO SCALE

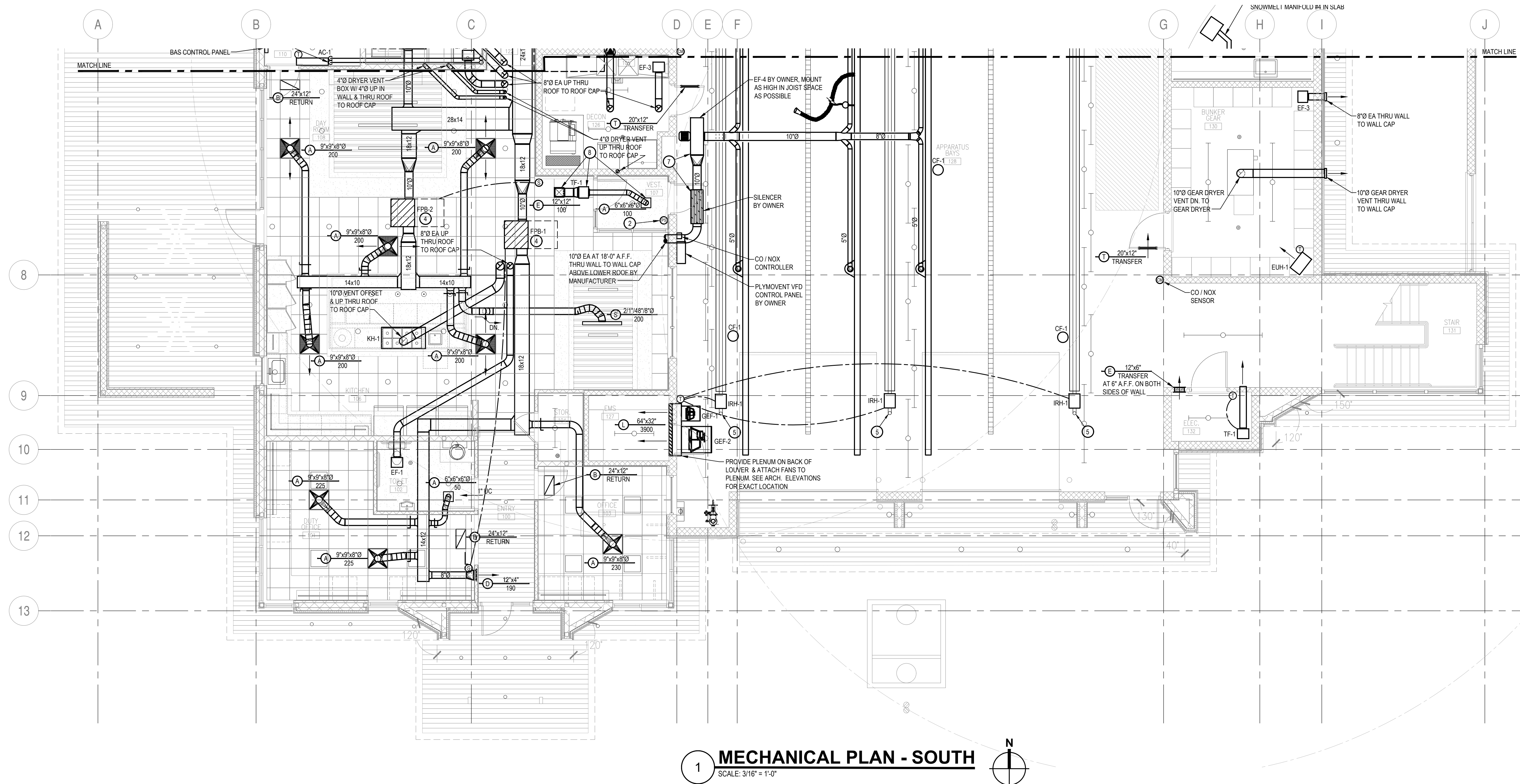


REVISION NAME	DATE

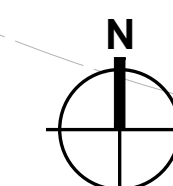
ISSUE DATE: 2024 0927

WORK NOTES:

- DIFFERENTIAL PRESSURE SENSOR ON WALL SET AT 0.05" W.G. (ADJ.) INTERLOCKED WITH FPB-3 TO ENERGIZE UNIT TO OPERATE IN RECIRCULATION MODE, IF NOT OPERATING WHEN THE SENSOR IS SATISFIED. IF PRESSURIZATION SENSOR IS SATISFIED THE UNIT SHALL RUN IN NORMAL OPERATING MODE TO MAINTAIN THERMOSTAT SETPOINT TEMPERATURE.
- DIFFERENTIAL PRESSURE SENSOR ON WALL SET AT 0.05" W.G. (ADJ.) INTERLOCKED WITH FPB2 TO ENERGIZE UNIT TO OPERATE IN RECIRCULATION MODE, IF NOT OPERATING WHEN THE SENSOR IS SATISFIED. IF PRESSURIZATION SENSOR IS SATISFIED THE UNIT SHALL RUN IN NORMAL OPERATING MODE TO MAINTAIN THERMOSTAT SETPOINT TEMPERATURE.
- NOT USED ON THIS SHEET.
- PROVIDE FAN POWER BOX (FPB) IDENTIFICATION BELOW FINISHED CEILING.
- 4"Ø INTAKE TO IRH THRU ROOF TO ROOF CAP.
- NOT USED ON THIS SHEET.
- INSTALLATION COORDINATION TO BY PROVIDED BY CONTRACTOR.
- 12x12 INTAKE GRILLE TO TRANSFER FAN WITH 6"Ø DISCHARGE TO SUPPLY DIFFUSER IN VESTIBULE AS SHOWN.
- NOT USED ON THIS SHEET.
- NOT USED ON THIS SHEET.



1 MECHANICAL PLAN - SOUTH
SCALE: 3/16" = 1'-0"



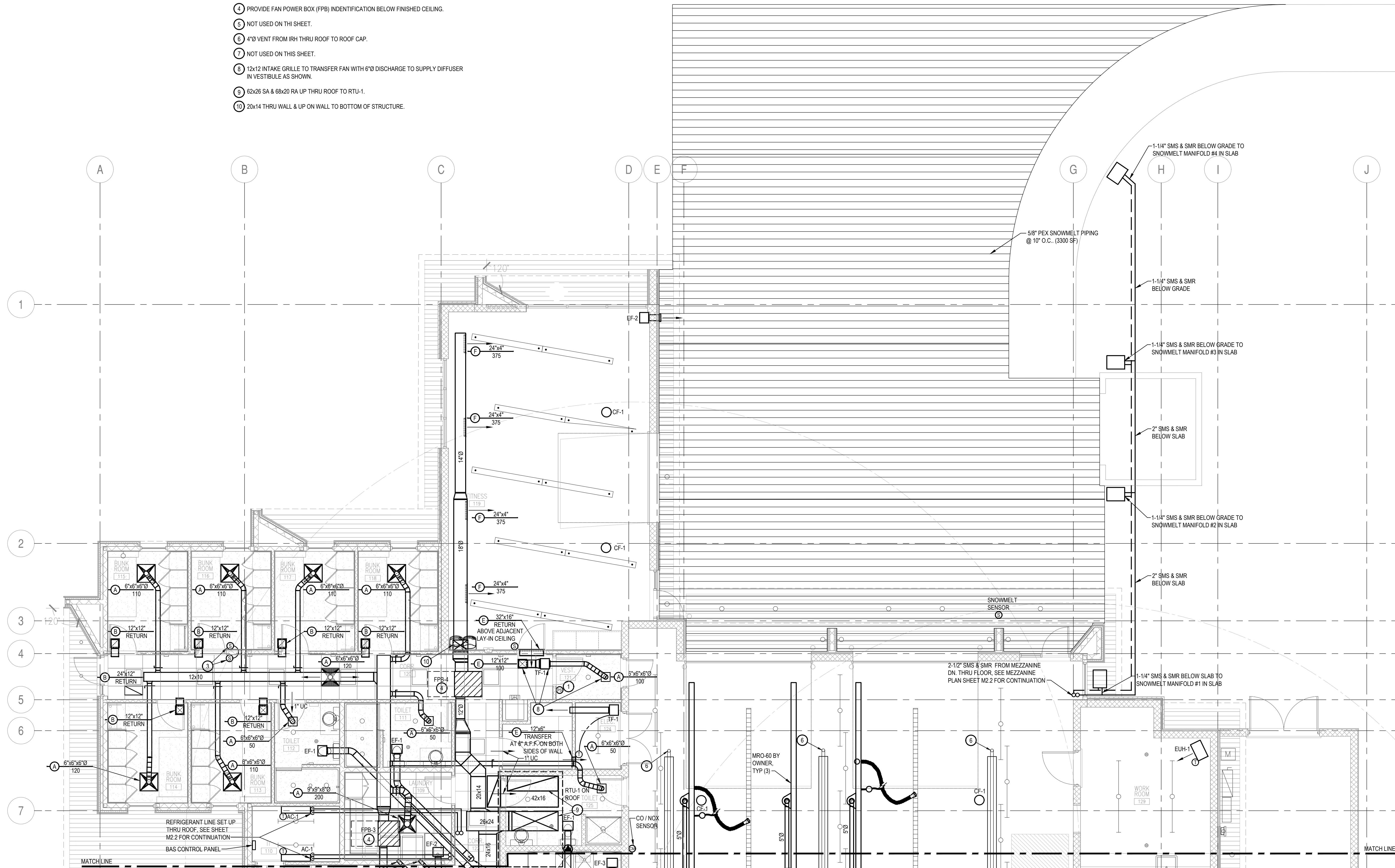


REVISION	NAME	DATE

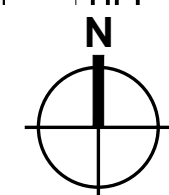
ISSUE DATE: 2024 0927

WORK NOTES:

- DIFFERENTIAL PRESSURE SENSOR ON WALL SET AT 0.05" W.C. (ADJ.) INTERLOCKED WITH FPB-3 TO ENERGIZE UNIT TO OPERATE IN RECIRCULATION MODE, IF NOT OPERATING WHEN THE SENSOR IS SATISFIED. IF PRESSURIZATION SENSOR IS SATISFIED THE UNIT SHALL RUN IN NORMAL OPERATING MODE TO MAINTAIN THERMOSTAT SETPOINT TEMPERATURE.
- DIFFERENTIAL PRESSURE SENSOR ON WALL SET AT 0.05" W.C. (ADJ.) INTERLOCKED WITH FPB2 TO ENERGIZE UNIT TO OPERATE IN RECIRCULATION MODE, IF NOT OPERATING WHEN THE SENSOR IS SATISFIED. IF PRESSURIZATION SENSOR IS SATISFIED THE UNIT SHALL RUN IN NORMAL OPERATING MODE TO MAINTAIN THERMOSTAT SETPOINT TEMPERATURE.
- PROVIDE SENSOR IN BUNK ROOM W/ REMOTE SENSOR LOCATED IN CORRIDOR. CLEARLY LABEL SENSOR SERVES BUNK ROOMS FPB-3.
- PROVIDE FAN POWER BOX (FPB) IDENTIFICATION BELOW FINISHED CEILING.
- NOT USED ON THI SHEET.
- 4"Ø VENT FROM IRH THRU ROOF TO ROOF CAP.
- NOT USED ON THIS SHEET.
- 12x12 INTAKE GRILLE TO TRANSFER FAN WITH 6"Ø DISCHARGE TO SUPPLY DIFFUSER IN VESTIBULE AS SHOWN.
- 62x26 SA & 68x20 RA UP THRU ROOF TO RTU-1.
- 20x14 THRU WALL & UP ON WALL TO BOTTOM OF STRUCTURE.



MECHANICAL PLAN - NORTH
SCALE: 3/16" = 1'-0"

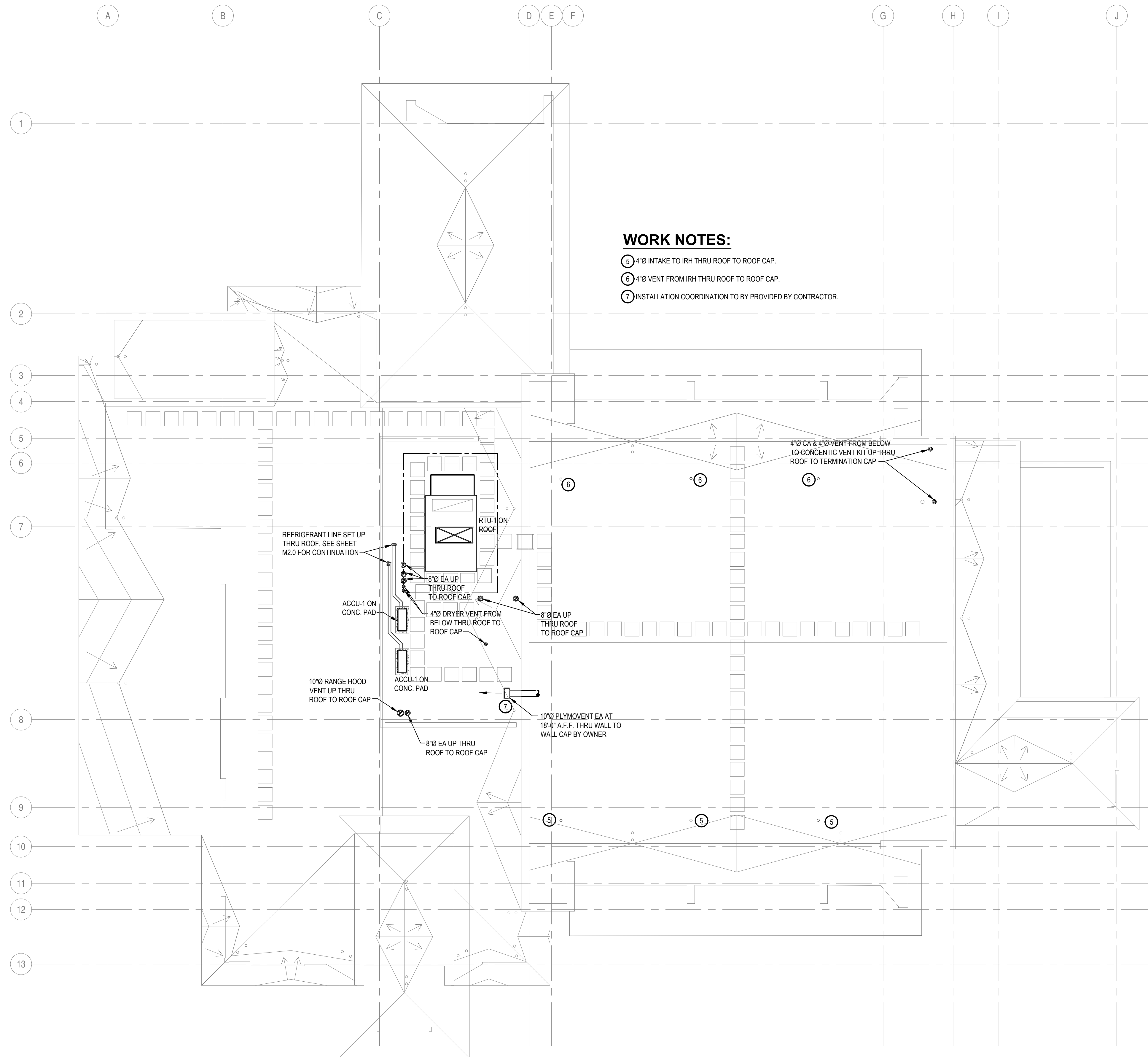


0 3' 6' 12'
SCALE: 3/16" = 1'-0"

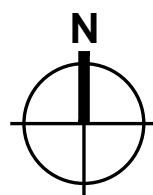


REVISION NAME	DATE

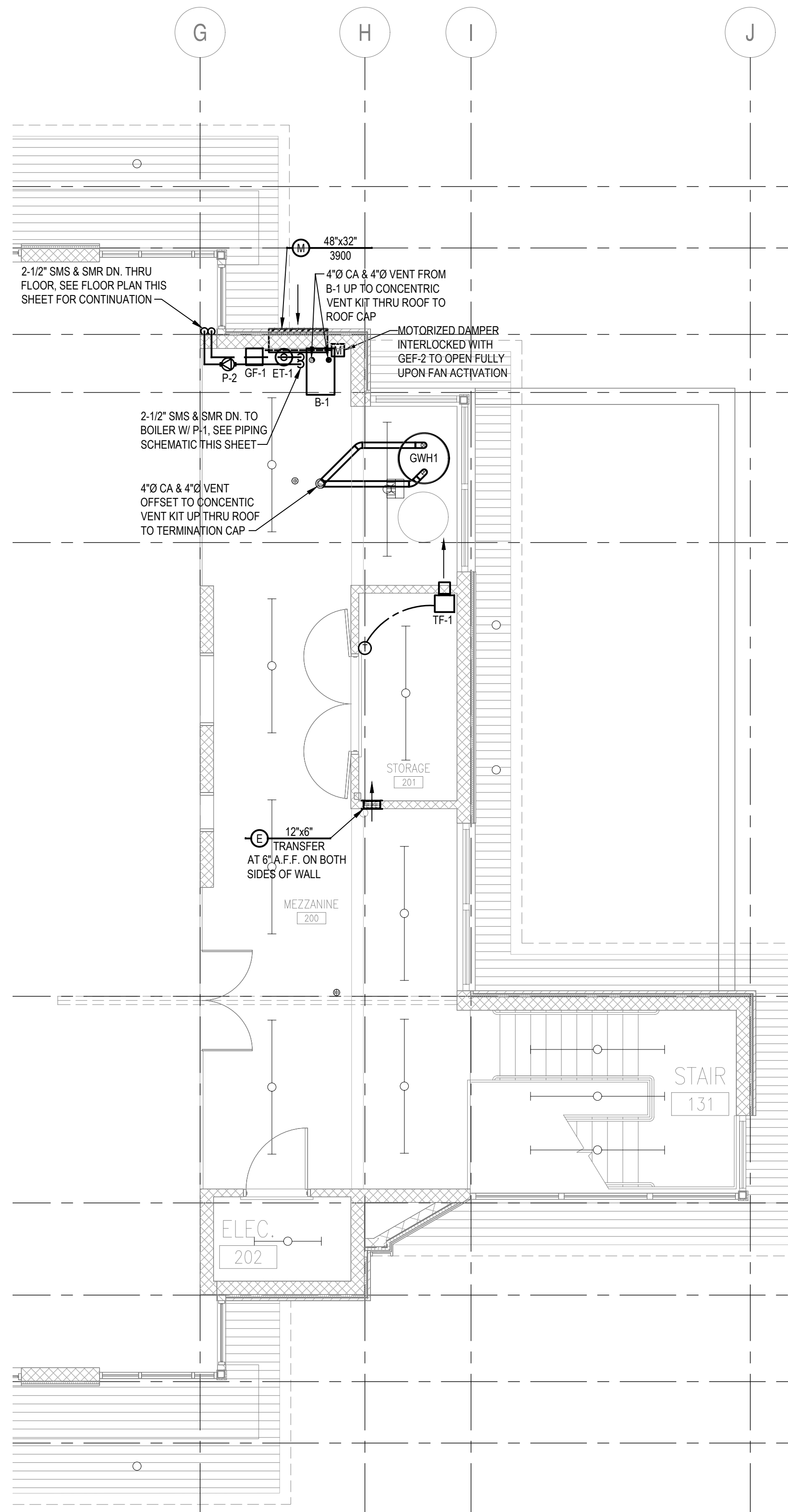
ISSUE DATE: 2024 0927



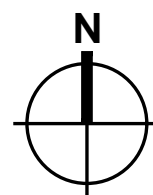
2 MECHANICAL ROOF PLAN
SCALE: 1/8" = 1'-0"



0 4' 8' 16'
SCALE: 1/8" = 1'-0"



1 MEZZANINE PLAN
SCALE: 3/16" = 1'-0"



0 3' 6' 12'
SCALE: 3/16" = 1'-0"



REVISION NAME	DATE

ISSUE DATE: 2024 0927

COMcheck Software Version COMcheckWeb Envelope Compliance Certificate

Project Information

Energy Code: 2021 IECC
Project Title: Thornton FS #8
Location: Thornton, Colorado
Climate Zone: 5b
Project Type: New Construction
Vertical Glazing / Wall Area: 30%

Construction Site: Monico St. & 153rd Ave
Thornton, Colorado
Owner/Agent: Rex Winquest
Dynamic MEP Consulting Engineer
1480 Hoyt St
Lakewood, Colorado 80215
rex@dmce.com

Designer/Contractor:

Additional Efficiency Package(s)

Credits: 10.0 Required 29.0 Proposed
Enhanced envelope performance, 7.0 credit
Reduced air infiltration, 4.0 credit
Reduced lighting power, 16.0 credit
Enhanced digital lighting controls, 2.0 credit

Building Area	Floor Area
1-Fire Station - Nonresidential	11091

Envelope Assemblies

Assembly	Gross Area or Perimeter	Cavity R-Value	Cont. R-Value	Proposed U-Factor	Budget U- Factor _{req}
Roof: Insulation Entirely Above Deck; [Bldg. Use 1 - Fire Station]	10324	---	34.2	0.029	0.032
NORTH					
Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Fire Station]	490	6.7	13.0	0.051	0.055
Window: Metal Frame with Thermal Break: Fixed, Perf. Specs.: Product ID N/A, SHGC 0.38, [Bldg. Use 1 - Fire Station] (b)	72	---	---	0.270	0.360
Ext. Wall: Concrete Block, 10in., Unreinforced, Cells Empty, Normal Density, Furring: Metal, [Bldg. Use 1 - Fire Station]	2213	0.0	13.0	0.061	0.090
Window: Metal Frame with Thermal Break: Fixed, Perf. Specs.: Product ID N/A, SHGC 0.38, [Bldg. Use 1 - Fire Station] (b)	728	---	---	0.270	0.360
Door: Insulated Metal, Non-Swinging, [Bldg. Use 1 - Fire Station]	112	---	---	0.200	0.179
EAST					
Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Fire Station]	125	6.7	13.0	0.051	0.055
Window: Metal Frame with Thermal Break: Fixed, Perf. Specs.: Product ID N/A, SHGC 0.38, [Bldg. Use 1 - Fire Station] (b)	56	---	---	0.270	0.360
Ext. Wall: Concrete Block, 10in., Unreinforced, Cells Empty, Normal Density, Furring: Metal, [Bldg. Use 1 - Fire Station]	2388	0.0	13.0	0.061	0.090
Window: Metal Frame with Thermal Break: Fixed, Perf. Specs.: Product ID N/A, SHGC 0.38, [Bldg. Use 1 - Fire Station] (b)	544	---	---	0.270	0.360
Door: Glass (over 50% glazing); Metal Frame, Entrance Door, Perf. Specs.: Product ID N/A, SHGC 0.51, [Bldg. Use 1 - Fire Station] (b)	114	---	---	0.290	0.630

Project Title: Thornton FS #8
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Assembly	Gross Area or Perimeter	Cavity R-Value	Cont. R-Value	Proposed U-Factor	Budget U- Factor _{req}
SOUTH					
Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Fire Station]	788	6.7	13.0	0.051	0.055
Window: Metal Frame with Thermal Break: Fixed, Perf. Specs.: Product ID N/A, SHGC 0.38, [Bldg. Use 1 - Fire Station] (b)	244	---	---	0.270	0.360
Ext. Wall: Concrete Block, 10in., Unreinforced, Cells Empty, Normal Density, Furring: Metal, [Bldg. Use 1 - Fire Station]	1904	0.0	13.0	0.061	0.090
Window: Metal Frame with Thermal Break: Fixed, Perf. Specs.: Product ID N/A, SHGC 0.80, [Bldg. Use 1 - Fire Station] (b)	841	---	---	0.270	0.360
Door: Insulated Metal, Non-Swinging, [Bldg. Use 1 - Fire Station]	112	---	---	0.200	0.179
WEST					
Ext. Wall: Steel-Framed, 16in. o.c., [Bldg. Use 1 - Fire Station]	1620	6.7	13.0	0.051	0.055
Window: Metal Frame with Thermal Break: Fixed, Perf. Specs.: Product ID N/A, SHGC 0.38, [Bldg. Use 1 - Fire Station] (b)	476	---	---	0.270	0.360
Ext. Wall: Concrete Block, 10in., Unreinforced, Cells Empty, Normal Density, Furring: Metal, [Bldg. Use 1 - Fire Station]	910	0.0	13.0	0.061	0.090

(a) Budget U-factors are used for software baseline calculations ONLY, and are not code requirements.
(b) Fenestration product performance must be certified in accordance with NFRC and requires supporting documentation.

Envelope PASSES: Design 23% better than code

Envelope Compliance Statement

Compliance Statement: The proposed envelope design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed envelope systems have been designed to meet the 2021 IECC requirements in COMcheck Version COMcheckWeb and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Name - Title Signature Date

Project Title: Thornton FS #8
Data filename: Report date: 06/13/24
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COMcheck Software Version COMcheckWeb Mechanical Compliance Certificate

Project Information

Energy Code: 2021 IECC
Project Title: Thornton FS #8
Location: Thornton, Colorado
Climate Zone: 5b
Project Type: New Construction

Construction Site: Monico St. & 153rd Ave
Thornton, Colorado
Owner/Agent: Rex Winquest
Dynamic MEP Consulting Engineer
1480 Hoyt St
Lakewood, Colorado 80215
rex@dmce.com

Designer/Contractor:

Additional Efficiency Package(s)

Credits: 10.0 Required 29.0 Proposed
Enhanced envelope performance, 7.0 credit
Reduced air infiltration, 4.0 credit
Reduced lighting power, 16.0 credit
Enhanced digital lighting controls, 2.0 credit

Mechanical Systems List

Quantity	System Type & Description
1	HVAC System (Multiple Zone): Heating: 1 each - Central Furnace, Gas, Capacity = 240 kBtu/h Proposed Efficiency = 81.00% Et, Required Efficiency: 81.00 % Et Cooling: 1 each - Single Package DX Unit, Capacity = 146 kBtu/h, Air-Cooled Condenser, Air Economizer Proposed Efficiency = 12.20 EER, Required Efficiency = 10.80 EER Proposed Part Load Efficiency = 14.00 IER, Required Part Load Efficiency = 14.00 IER Fan System: FAN SYSTEM 1 - Compliance (Brake HP and fan efficiency method) - Passes Fans: FAN 1 Supply, Multi-Zone VAV, 4800 CFM, 5.0 motor nameplate hp, 3.7 design brake hp (3.7 max. BHP), 1.10 fan energy index
2	HVAC System (Single Zone): Cooling: 2 each - Split System, Capacity = 18 kBtu/h, Air-Cooled Condenser, Unknown Economizer Proposed Efficiency = 16.00 SEER2, Required Efficiency = 13.40 SEER2 Proposed Part Load Efficiency = 0.00, Required Part Load Efficiency = 0.00
1	Water Heater: Gas Storage Water Heater, Capacity: 100 gallons, Input Rating: 250 kBtu/h w/ Circulation Pump Proposed Efficiency: 96.00 % Et, Required Efficiency: 80.00 % Et

Mechanical Compliance Statement

Compliance Statement: The proposed mechanical design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed mechanical systems have been designed to meet the 2021 IECC requirements in COMcheck Version COMcheckWeb and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Name - Title Signature Date

Project Title: Thornton FS #8
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COMcheck Software Version COMcheckWeb Inspection Checklist

Energy Code: 2021 IECC

Requirements: 0.0% were addressed directly in the COMcheck software

Text in the "Comments/Assumptions" column is provided by the user in the COMcheck Requirements screen. For each requirement, the user certifies that a code requirement will be met and how that is documented, or that an exception is being claimed. Where compliance is itemized in a separate table, a reference to that table is provided.

Section # & Req-ID	Plan Review	Complies?	Comments/Assumptions
C103.2 [PR1] ¹	Plans and/or specifications provide all information with which compliance can be determined for the building envelope and document where exceptions to the standard are claimed.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C103.2 [PR2] ¹	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the mechanical and service water heating systems and document where exceptions to the standard are claimed. Load calculations per acceptable engineering standards and handbooks. Hot water system sized per manufacturer's sizing guide.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C103.2 [PR4] ¹	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the interior lighting and electrical systems and equipment and document where exceptions to the standard are claimed. Information provided should include interior lighting power calculations, wattage of bulbs and ballasts, transformers and control devices.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C103.2 [PR8] ¹	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the exterior lighting and electrical systems and equipment and document where exceptions to the standard are claimed. Information provided should include exterior lighting power calculations, wattage of bulbs and ballasts, transformers and control devices.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.4.1 [PR10] ¹	The vertical fenestration area <= 30 percent of the gross above-grade wall area.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.4.1 [PR11] ¹	The skylight area <= 3 percent of the gross roof area.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

Project Title: Thornton FS #8
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Section # & Req-ID	Plan Review	Complies?	Comments/Assumptions
C402.4.2 [PR14] ¹	In enclosed spaces > 2,500 ft ² directly under a roof with ceiling heights >15 ft. and used as an office, lobby, atrium, concourse, corridor, storage, gymnasium/exercise center, convention center, automotive service, manufacturing, non-refrigerated warehouse, retail store, distribution/sorting area, transportation, or workshop, the following requirements apply: (a) the daylight zone under skylights is >= half the floor area; (b) the skylight area to daylight zone is >= 3 percent with a skylight VT >= 0.40; or a minimum skylight effective aperture >= 1 percent.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C406 [PR9] ¹	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the additional energy efficiency package options.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

Project Title: Thornton FS #8
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Section # & Req-ID	Footing / Foundation Inspection	Complies?	Comments/Assumptions
C303.2.1 [FO6] ¹	Exterior insulation protected against damage, sunlight, moisture, wind, landscaping and equipment maintenance activities.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.13.2 [FO9] ¹	Snow/ice melting system and freeze protection systems have sensors and controls configured to limit service for pavement temperature above 50F and outdoor temperature above 40F.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

Project Title: Thornton FS #8
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Section # & Req-ID	Framing / Rough-In Inspection	Complies?	Comments/Assumptions
C303.1.3 [FR12] ¹	Fenestration products rated in accordance with NFRC certified and as to performance labels or certificates provided.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.4.3 [FR10] ¹	Vertical fenestration SHGC value.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Envelope Assemblies table for values.
C402.4.3.4 [FR8] ¹	Installed vertical fenestration U-factor and SHGC consistent with label specifications and as reported in plans and COMcheck reports.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Envelope Assemblies table for values.
C402.4.5 [FR14] ¹	U-factor of opaque swinging and nonswinging doors associated with the building thermal envelope meets requirements.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Envelope Assemblies table for values.
C402.5.1.3 [FR19] ¹	The building envelope contains a continuous air barrier that is sealed in an approved manner and material permeability <= 0.004 dfm/ft ² . Air barrier penetrations are sealed in an approved manner.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.5.4 [FR18] ¹	Factory-built fenestration and doors are labeled as meeting air leakage requirements.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

Project Title: Thornton FS #8
Data filename: Report date: 06/13/24
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Section # & Req-ID	Plumbing Rough-In Inspection	Complies?	Comments/Assumptions
C404.5.1 [PL6] ¹	Heated water supply piping conforms to pipe length and volume requirements. Refer to section details.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C404.6.1.1 [PL3] ¹	Automatic time switches installed to automatically switch off the recirculating hot-water system or heat trace.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C404.6.3 [PL7] ¹	Pumps that circulate water between a heater and storage tank have controls that limit operation from startup to <= 5 minutes after end of heating cycle.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C404.6.1.1 [PL8] ¹	Demand recirculation water systems have controls that start the pump upon receiving a signal from the action of a user of a fixture or appliance and limits the temperature of the water entering the cold-water piping to 104°F.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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REVISION NAME	DATE

ISSUE DATE: 2024 0927

Section # & Req.ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C402.2.6 [ME41] ¹	Thermally ineffective panel surfaces of sensible heating panels have insulation $\geq$ R-3.5.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.7.7 [ME58] ¹	Outdoor air and exhaust systems have motorized dampers that automatically shut when not in use and meet maximum leakage rates. Check gravity dampers where allowed. Reference section language for operational details.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.8.1 [ME65] ¹	HVAC fan systems at design conditions do not exceed allowable fan system motor nameplate hp or fan system bhp.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Mechanical Systems list for values.
C403.8.3 [ME117] ¹	Fans have a fan energy index (FEI) $\geq$ 1.00. Variable volume fans will have an FEI $\geq$ 0.95.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.8.4 [ME142] ¹	Motors for fans that are not less than 1/12 hp and less than 1 hp are electronically commutated motors or have a minimum motor efficiency of 70 percent. These motors have the means to adjust motor speed.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.8.6 [ME143] ¹	Each DX cooling system > 65 kBTu and chiller water/evaporative cooling system with fans $\geq$ 1/4 hp are designed to vary the indoor fan airflow as a function of load and comply with detailed requirements of this section.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.9 [ME144] ¹	Large diameter fans where installed shall be tested and labeled in accordance with AMCA 230.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.3 [ME55] ¹	HVAC equipment efficiency verified.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Mechanical Systems list for values.
C403.2.1 [ME112] ¹	Zone isolation devices and controls installed where applicable.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.5.5 [ME113] ¹	Fault detection and diagnostics installed with air-cooled unitary DX units or VRF units having economizers.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.2.2 [ME59] ¹	Natural or mechanical ventilation is provided in accordance with International Mechanical Code Chapter 4. Mechanical ventilation has capability to reduce outdoor air supply to minimum per IMC Chapter 4.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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Section # & Req.ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C403.6.1 [ME75] ¹	Hydronic and multizone HVAC system controls are VAV fans driven by mechanical or electrical variable speed drive per Table C403.4.1.1.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.9 [ME67] ¹	VAV fans have static pressure sensors located so controller setpoint $\leq$ 1.2 w.c..	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.4.1.3 [ME24] ¹	Reset static pressure setpoint for DDC controlled VAV boxes reporting to central controller based on the zones requiring the most pressure.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.1 [ME130] ¹	Supply air systems serving multiple zones have VAV systems with controls configured to reduce the volume of air that is reheated, re-cooled or mixed in each zone. See section for details.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.2 [ME131] ¹	Single-duct VAV systems use terminal devices configured to reduce the supply of primary supply air before reheating or recooling takes place.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.3 [ME132] ¹	Systems that have 1 warm air duct and 1 cool air duct use terminal devices configured to reduce the flow from one duct to a minimum before mixing of air from the other duct takes place.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.4 [ME133] ¹	Individual dual-duct or mixing heating and cooling systems with a single fan with total capacities $\geq$ 90,000 Btu/h not equipped with air economizers.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.5 [ME134] ¹	Multiple zone HVAC systems have supply air temperature reset controls based on building loads or outside temperatures.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.7 [ME136] ¹	Parallel flow fan-powered VAV air terminals have automatic controls configured to 1) turn off the terminal fan except when space heating is required or where required for ventilation, 2) turn on the terminal fan as the first stage of heating before the heating coil is activated, and 3) during heating for warmup or setback temperature control, either operate the terminal fan and heating coil without primary air or, reverse the terminal damper logic and provide heating from the central air handler by primary air.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.8 [ME137] ¹	Systems with DDC of individual zones reporting to the central control panel configured to reset the static pressure setpoint based on zone requiring the most pressure. The DDC is capable of monitoring zone damper positions or have an alternative method of indicating the need for static pressure. See section for details.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

Project Title: Thornton FS #8 Report date: 06/13/24
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Section # & Req.ID	Rough-In Electrical Inspection	Complies?	Comments/Assumptions
C405.2.3.1 [EL22] ¹	Spaces required to have light reduction controls have a manual control that allows the occupant to reduce the connected lighting load in a reasonably uniform illumination pattern $\geq$ 50 percent.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.2.1.1 [EL18] ¹	Occupancy sensors installed in classrooms/lecture/training rooms, conference/meeting/multipurpose rooms, copy/print rooms, lounges/breakrooms, enclosed offices, open plan office areas, restrooms, storage rooms, locker rooms, corridors, warehouse storage areas, and other spaces $\leq$ 300 sqft that are enclosed by floor-to-ceiling height partitions. Reference section language C405.2.1.2 for control function in warehouses and section C405.2.1.3 for open plan office spaces.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.2.1.2 [EL19] ¹	Occupancy sensors control function in warehouses. In warehouses, the lighting in aislemways and open areas is controlled with occupancy sensors that automatically reduce lighting power by 50% or more within 20 minutes of when the areas are unoccupied. The occupant sensors control lighting in each aisleway independently and do not control lighting beyond the aisleway being controlled by the sensor. Lights not turned off by occupant sensors is done so by time-switch.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.2.1.3 [EL20] ¹	Occupant sensor control function in open plan office areas: Occupant sensor controls in open office spaces $\geq$ 300 sqft. have controls 1) configured so that general lighting can be controlled separately in control zones with floor areas $\leq$ 600 sq ft. within the space, 2) general lighting in each zone permitted to turn on upon occupancy in control zone, 3) automatically turn off general lighting in all control zones within 20 minutes after all occupants have left the space, 4) are configured so that general lighting power in each control zone is reduced by $\geq$ 80% of the full zone general lighting power within 20 minutes of all occupants leaving that control zone.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.2.2.1 [EL21] ¹	Each area not served by occupancy sensors (per C405.2.1.1) have time-switch controls and functions detailed in sections C405.2.2.1.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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Section # & Req.ID	Insulation Inspection	Complies?	Comments/Assumptions
C303.1 [IN3] ¹	Roof insulation installed per manufacturer's instructions and is labeled with R-value or insulation certificate providing R-value and other relevant data. Blown or poured loose-fill insulation is installed only where the roof slope is $\leq$ 3 in 12.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.2.1 [IN2] ¹	Roof assembly meets minimal thermal resistance installed between roof framing or in a continuous fashion on the roof assembly as stipulated in Table C402.1.3. Requirements for above deck insulation, minimum thickness, suspended ceilings, staggered joints and skylight curbs will be met.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C303.2 [IN7] ¹	Above-grade wall insulation installed per manufacturer's instructions.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.1.3 [IN12] ¹	Non-swinging opaque doors have R-4.75 insulation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C105 [IN6] ¹	Installed above-grade wall insulation type and R-value consistent with insulation specifications reported in plans and COMcheck reports.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Envelope Assemblies table for values.
C402.2.6 [IN18] ¹	Radiant panels and associated components, designed for heat transfer from the panel surfaces to the occupants or indoor space are insulated with a minimum of R-3.5.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Envelope Assemblies table for values.
C105 [IN2] ¹	Installed roof insulation type and R-value consistent with insulation specifications reported in plans and COMcheck reports. For some ceiling systems, verification may need to occur during Framing Inspection.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.5.1.1 [IN1] ¹	All sources of air leakage in the building thermal envelope are sealed, caulked, gasketed, weather stripped or wrapped with moisture vapor-permeable wrapping material to minimize air leakage.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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Section # & Req.ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C403.7.1 [ME59] ¹	Demand control ventilation provided for spaces $\geq$ 500 ft ² and $\geq$ 15 people/1,000 ft ² occupant density and served by systems with air side economizer, auto modulating outside air damper control, or design airflow $\geq$ 3,000 cfm.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.7.2 [ME115] ¹	Enclosed parking garage ventilation has automatic contaminant detection and capacity to stage or modulate fans to 50% or less of design capacity.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.7.3 [ME140] ¹	Units that provide ventilation air to multiple zones and operate in combination with zone heating and cooling systems do not use heating or heat recovery to warm supply air to a temperature greater than 60°F when representative building loads or outdoor air temperatures indicate that the majority of zones require cooling.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.7.6 [ME141] ¹	HVAC systems serving guestrooms in Group R-1 buildings with $>$ 50 guestrooms: Each guestroom is provided with controls that automatically manage temperature setpoint and ventilation (see sections C403.7.6.1 and C403.7.6.2).	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.7.4 [ME57] ¹	Exhaust air energy recovery on systems meeting Table C403.7.4(1) and C403.7.4(2).	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.7.5 [ME116] ¹	Kitchen exhaust systems comply with replacement air and conditioned supply air limitations, and satisfy hood rating requirements and minimum exhaust rate criteria.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.5.1, C403.5.2 [ME62] ¹	Air economizers provided where required, meet the requirements for design capacity, control signal, ventilation controls, high-limit shut-off, integrated economizer control, and provide a means to relieve excess outside air during operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.5.3.3 [ME124] ¹	Air economizers automatically reduce outdoor air intake to the design minimum outdoor air quantity when outdoor air intake will not reduce cooling energy usage. See Table C403.5.3.3 for applicable device types and climate zones.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.5.3.4 [ME125] ¹	System capable of relieving excess outdoor air during air economizer operation to prevent over pressurizing the building. The relief air outlet located to avoid recirculation into the building.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.5.3.5 [ME126] ¹	Return, exhaust/relief and outdoor air dampers used in economizers have motorized dampers that automatically shut when not in use and meet maximum leakage rates. Reference section C403.7.7 for details.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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Section # & Req.ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C403.6.9 [ME138] ¹	Static pressure sensors used to control VAV fans located such that the controller setpoint is $\leq$ 1.2 inches w.c.. Where this results in one or more sensors being located downstream of major duct splits, not less than one sensor located on each major branch.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.4.1.4 [ME63] ¹	Heating for vestibules and air curtains with integral heating include automatic controls that shut off the heating system when outdoor air temperatures $>$ 45°F. Vestibule heating and cooling systems controlled by a thermostat in the vestibule with heating setpoint $\leq$ 60°F and cooling setpoint $\geq$ 60°F.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.6.6 [ME135] ¹	Multiple zone VAV systems with DDC of individual zone boxes have static pressure setpoint reset controls.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Mechanical Systems list for values.
C403.3.3 [ME35] ¹	Hot gas bypass limited to: $\leq$ 240 kBTu/h - 50% $>$ 240 kBTu/h - 25%	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.2.1 [ME111] ¹	Gas-fired water-heating equipment installed in new buildings where a singular piece of water-heating equipment $\geq$ 1,000 kBTu/h serves the entire building, thermal efficiency $\geq$ 92 Et. Where multiple pieces of water-heating equipment serve the building with combined rating $\geq$ 1,000 kBTu/h, the combined input capacity weighted-average thermal efficiency $\geq$ 90 Et. Exclude input rating of equipment in individual dwelling units and equipment $\leq$ 100 kBTu/h.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.2.1 [ME53] ¹	Air outlets and zone terminal devices have means for air balancing.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.11.3.1, C403.11.3.2 [ME123] ¹	Refrigerated display cases, walk-in coolers or walk-in freezers served by remote compressors and remote condensers not located in a condensing unit, have fan-powered condensers that comply with Sections C403.11.3.1 and refrigeration compressor systems that comply with C403.11.3.2.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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Section # & Req.ID	Rough-In Electrical Inspection	Complies?	Comments/Assumptions
C405.2.4.1, C405.2.4.2 [EL23] ¹	Daylight zones provided with individual controls that control the lights independent of general area lighting. See code section C405.2.3. Daylight-responsive controls for applicable spaces, C405.2.3.1. Daylight responsive control function and section C405.2.3.2. Sidelight zone.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.2.5 [EL27] ¹	Additional interior lighting power allowed for special functions per the approved lighting plans and is automatically controlled and separated from general lighting.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.2.7 [EL28] ¹	Automatic lighting controls for exterior lighting installed. Controls will be daylight controlled, set based on business operation time-of-day, or reverse connected lighting $>$ 30%.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.7 [EL26] ¹	Low-voltage dry-type distribution electric transformers meet the minimum efficiency requirements of Table C405.6.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.8 [EL27] ¹	Electric motors meet the minimum efficiency requirements of Tables C405.7(1) through C405.7(4). Efficiency verified through certification under an approved certification program or the equipment efficiency ratings shall be provided by motor manufacturer (where certification programs do not exist).	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.9.1, C405.9.2 [EL28] ¹	Escalators and moving walks comply with ASME A17.1/CSA B44 and have automatic controls configured to reduce speed to the minimum permitted speed in accordance with ASME A17.1/CSA B44 or applicable local code when not conveying passengers.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.10 [EL29] ¹	Total voltage drop across the combination of feeders and branch circuits $\leq$ 5%.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.1.1 [EL30] ¹	At least 90% of dwelling unit permanently installed lighting shall have lamp efficacy $\geq$ 65 lm/W or luminaires with efficacy $\geq$ 45 lm/W or comply with C405.2.4 or C405.3.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.11, C405.11.1 [EL31] ¹	50% of 15/20 amp receptacles installed in enclosed offices, conference rooms, copy rooms, break rooms, classrooms and workstations and $>$ 25% of branch circuit feeders for modular furniture will have automatic receptacle control in accordance with C405.11.1.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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Section # & Req.ID	Final Inspection	Complies?	Comments/Assumptions
C303.3, C408.2.5.2 [F117] ¹	Furnished O&M instructions for systems and equipment to the building owner or designated representative.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C303.3, C408.2.5.3 [F18] ¹	Furnished O&M manuals for HVAC systems within 90 days of system acceptance.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C401.3 [F158] ¹	A thermal envelope certificate will be supplied and completed by an approved third party.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.5.10 [F126] ¹	Recessed luminaires in thermal envelope to limit infiltration and be labeled and labeled. Seal between interior finish and luminaire housing.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.5.11 [F159] ¹	Operable openings $>$ 40 ft ² will be interlocked with heating and cooling systems to setback setpoint temperatures within 10 minutes of opening.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C402.5.8 [F137] ¹	Weatherstrials installed on all loading dock cargo door openings and provide direct contact along the top and sides of vehicles parked in the doorway.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.3.1 [F127] ¹	HVAC systems and equipment capacity does not exceed calculated loads.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.4.1 [F147] ¹	Heating and cooling to each zone is controlled by a thermostat control. Minimum one humidity control device per installed humidification/dehumidification system.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.4.1.2 [F138] ¹	Thermostatic controls have a 5 °F deadband.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.4.1.3 [F120] ¹	Temperature controls have setpoint overlap restrictions.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.4.2 [F139] ¹	Each zone equipped with setback controls using automatic time clock or programmable control system.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C403.4.2.1, C403.4.2.2 [F140] ¹	Automatic Controls: Setback to 55°F (heat) and 65°F (cool); 7-day clock, 2-hour occupant override, 10-hour backup.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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Section # & Req.ID	Final Inspection	Complies?	Comments/Assumptions
C403.4.2.3 (F141) ¹	Systems include optimum start controls.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C404.3 (F111) ¹	Heat traps installed on supply and discharge piping of non-circulating systems.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C404.4 (F125) ¹	All piping insulated in accordance with section details and Table C403.12.3.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C404.6.1 (F112) ¹	Controls are installed that limit the operation of a recirculation pump installed to maintain temperature of a storage tank. System return pipe is a dedicated return pipe or a cold water supply pipe.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C405.5.1 (F119) ¹	Exterior lighting power is consistent with what is shown on the approved lighting plans, demonstrating proposed watts are less than or equal to allowed watts.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Exterior Lighting fixture schedule for values.
C406.3 (F167) ¹	Reduced lighting power - this credit specifies that the connected lighting power is >= 10% more efficient than 2021 IECC requirements.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C406.4 (F154) ¹	Enhanced Digital Lighting Controls - Interior lighting has the following enhanced lighting controls in accordance with Sections C405.2.1 through C405.2.3. Luminaires capable of continuous dimming and being addressed individually, at least 8 luminaires controlled in combination in a daylight zone, digital control system for fixtures with load shedding or occupancy sensors. Sequence of Operations documentation, and functional testing per Section C406.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C406.8 (F168) ¹	Enhanced envelope performance - the building thermal envelope UA value is >= 15% better than the total UA of the envelope specified by Section C402.1.5.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C406.9 (F156) ¹	Reduced air infiltration energy - the measured air-leakage rate of the building envelope is lower than 0.25 cfm/ft2. Comprehensive report documentation will be submitted to the code official and the building owner.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.1.1 (F157) ¹	Building operations and maintenance documents will be provided to the owner. Documents will cover manufacturers' information, specifications, programming procedures and means of illustrating to owner how building, equipment and systems are intended to be installed, maintained, and operated.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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Section # & Req.ID	Final Inspection	Complies?	Comments/Assumptions
C408.2.1 (F128) ¹	Commissioning plan developed by registered design, professional or approved agency.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.3.1 (F131) ¹	HVAC equipment, systems and system-to-system relationships have been tested to ensure proper operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.3.2 (F110) ¹	HVAC and service water heating control systems have been tested to ensure proper operation, calibration and adjustment of controls.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.3.3 (F132) ¹	Economizers have been tested to ensure proper operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.4 (F129) ¹	Preliminary commissioning report completed and certified by registered design professional or approved agency.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.5 (F17) ¹	Furnished HVAC as-built drawings submitted within 90 days of system acceptance.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.5 (F116) ¹	Furnished as-built drawings for electric power systems within 90 days of system acceptance.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.5.1 (F143) ¹	An air and/or hydronic system balancing report is provided for HVAC systems.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.2.5.2 (F130) ¹	Final commissioning report due to building owner within 90 days of receipt of certificate of occupancy.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	
C408.3 (F133) ¹	Lighting systems have been tested to ensure proper calibration, adjustment, programming, and operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

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